Renal Reference — FENa Interpretation

▼ When FeNa is "best-case" (highest yield)

FeNa is **most useful** when ALL are true:

- **Oliguric AKI**
- **Not on diuretics**
- **No CKD**

This is when FeNa best separates **prerenal (low FeNa)** vs **ATN (high FeNa)**.

▼ When FeNa gets weaker (why + what to do)

A) Nonoliguric AKI

- FeNa becomes **less reliable**
- Reason: many intrinsic AKI states are **nonoliguric** and don't follow the classic pattern
- Classic exam example: **aminoglycoside toxicity → nonoliguric AKI**, so FeNa may not cleanly separate prerenal vs ATN

B) CKD

- FeNa is **less reliable**
- Reason: CKD tubules may not conserve sodium normally
- Result: **prerenal + CKD may have FeNa not low** → misleading

C) Diuretics

- FeNa becomes **unreliable** because diuretics force sodium into urine → **falsely high FeNa**
- What to use instead: **FeUrea**
  - **FeUrea <35%** → supports **prerenal**
  - **FeUrea >50%** → supports **intrinsic AKI (ATN)**

The single most important sentence

FeNa is not useless outside the ideal setting — it's just much less trustworthy.

Use it as **supporting evidence**, not the final answer.

1. FENa formula

WHAT YOU SEE

FENa equation (added reference)

WHAT IT ENCODES

FENa = (UNa × PCr) / (PNa × UCr) × 100, expressed as a percent. It is the fraction of filtered sodium that the kidney excretes rather than reabsorbs. The plasma and urine creatinine terms normalize for how much filtrate was made, so FENa is far more reliable than a spot urine Na alone (a low urine Na can be from low filtration, not avid reabsorption).

BOARD TRAP

A low spot urine sodium alone proves prerenal AKI. WRONG — urine Na is confounded by urine concentration; you must use FENa (or FEUrea) which corrects for the creatinine/filtration term.

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2. Best-case header

WHAT YOU SEE

Top green box header 'When FENa is best-case (highest yield)'

WHAT IT ENCODES

FENa is most trustworthy only when three conditions hold: the AKI is OLIGURIC (kidney is actively deciding how to handle Na), the patient is NOT on diuretics, and there is no CKD. Outside these conditions FENa loses discriminating power and you fall back on history, urine sediment, and FEUrea.

BOARD TRAP

FENa is a valid first-line discriminator in every AKI workup. WRONG — it is only trustworthy in oliguric AKI, off diuretics, without CKD; outside those conditions (the common ward scenario) it misleads and you must use FEUrea plus sediment.

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3. Oliguric AKI

WHAT YOU SEE

Bullet 'Oliguric AKI'

WHAT IT ENCODES

FENa performs best in OLIGURIC AKI (urine output <400–500 mL/day) because the tubules are making a strong sodium-handling decision — conserve hard (prerenal, FENa <1%) or fail to reabsorb (ATN, FENa >2%). In nonoliguric AKI the signal is muddier.

BOARD TRAP

FENa is equally valid in nonoliguric AKI. WRONG — nonoliguric ATN (classically aminoglycoside, cisplatin, or contrast) can show FENa that does not cleanly separate from prerenal; rely on sediment and context instead.

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4. Not on diuretics

WHAT YOU SEE

Bullet 'Not on diuretics'

WHAT IT ENCODES

Loop and thiazide diuretics block tubular Na reabsorption and dump Na into the urine → FENa is FALSELY HIGH even in a truly volume-depleted prerenal patient. FENa is only interpretable if the patient is off diuretics for roughly 24h; otherwise switch to FEUrea (urea handling is less affected by loop/thiazide agents).

BOARD TRAP

A FENa >2% in a patient on furosemide confirms ATN. WRONG — the diuretic itself forces natriuresis and inflates FENa; a volume-depleted prerenal patient on a loop diuretic can read >2%. Use FEUrea <35% to unmask the prerenal state.

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5. No CKD

WHAT YOU SEE

Bullet 'No CKD'

WHAT IT ENCODES

Established CKD damages tubular sodium-conserving capacity, so a CKD patient may run a baseline FENa >1% even when prerenal/volume-depleted. Chronic diuretic use, salt-wasting nephropathy, and adrenal insufficiency similarly raise baseline FENa. Interpret FENa against the patient's known baseline kidney function.

BOARD TRAP

A CKD patient with FENa 1.5% during an acute illness must have ATN. WRONG — diseased CKD tubules cannot conserve Na maximally, so prerenal-on-CKD can read >1%; correlate with volume status and trend, not the single FENa value.

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6. Separates prerenal vs ATN

WHAT YOU SEE

Line 'FENa best separates prerenal (low FENa) vs ATN (high FENa)'

WHAT IT ENCODES

Core use: FENa <1% = prerenal (avid Na reabsorption from intact tubules responding to hypoperfusion); FENa >2% = ATN (injured tubules cannot reabsorb Na). The 1–2% gray zone is indeterminate. Prerenal is usually accompanied by urine osm >500, urine Na <20, BUN:Cr >20, and bland sediment; ATN by urine osm ~300 (isosthenuria), urine Na >40, and muddy-brown granular casts.

BOARD TRAP

FENa <1% always means prerenal. WRONG — FENa is <1% in several INTRINSIC conditions: early sepsis, contrast-induced AKI, rhabdomyolysis/pigment nephropathy, acute glomerulonephritis, and hepatorenal syndrome. These keep tubular Na avidity intact despite real parenchymal/intrinsic injury.

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7. When FENa gets weaker

WHAT YOU SEE

Pink box header 'When FENa gets weaker (why + what to do)'

WHAT IT ENCODES

Three scenarios degrade FENa reliability: nonoliguric AKI, underlying CKD, and active diuretic use. The fix is to (a) recognize the limitation, (b) substitute FEUrea when diuretics are on board, and (c) lean on urine sediment, history, and fluid-challenge response.

BOARD TRAP

A FENa in the 1–2% gray zone gives a definite answer. WRONG — the 1–2% band is indeterminate, and nonoliguric AKI, CKD, and diuretics all push FENa into or above this range; treat a gray-zone FENa as non-diagnostic and use FEUrea/sediment.

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8. A) Nonoliguric AKI

WHAT YOU SEE

Section A 'Nonoliguric AKI'

WHAT IT ENCODES

Many intrinsic AKIs are NONoliguric (urine output preserved), so the tubules aren't conserving Na maximally and FENa loses its clean cut-point. Nonoliguric ATN actually carries a better prognosis than oliguric ATN but is harder to classify by FENa — use sediment (muddy-brown casts) instead.

BOARD TRAP

Preserved urine output in AKI means the kidney injury is mild or prerenal. WRONG — nonoliguric ATN (aminoglycoside, cisplatin, contrast) maintains urine output yet is true tubular injury; muddy-brown casts confirm it despite a non-diagnostic FENa.

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9. Aminoglycoside example

WHAT YOU SEE

Line 'aminoglycoside toxicity → nonoliguric AKI'

WHAT IT ENCODES

Aminoglycoside (and cisplatin) nephrotoxicity classically causes NONoliguric ATN with associated tubular wasting (hypomagnesemia, hypokalemia) → FENa may not separate prerenal from ATN. Diagnosis rests on drug exposure timeline (5–10 days), maintained urine output, and granular casts rather than on FENa.

BOARD TRAP

A nonoliguric patient with a borderline FENa on day 7 of gentamicin is prerenal. WRONG — aminoglycoside ATN is the classic nonoliguric nephrotoxic injury; preserved urine output and a FENa near 1% do not exclude it. Check the drug course and urine sediment.

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10. B) CKD

WHAT YOU SEE

Section B 'CKD'

WHAT IT ENCODES

In CKD the tubules have lost maximal Na-conserving ability, so a prerenal insult superimposed on CKD often gives FENa >1% — making FENa look like ATN when it isn't. Trend the creatinine from baseline and assess volume; a fluid challenge response is more informative than a single FENa.

BOARD TRAP

A CKD patient with FENa >1% during an acute drop in function has ATN, not prerenal. WRONG — diseased CKD tubules cannot conserve Na maximally, so prerenal-on-CKD reads >1%; trend the creatinine from baseline and test fluid responsiveness instead of trusting the FENa.

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11. C) Diuretics

WHAT YOU SEE

Section C 'Diuretics'

WHAT IT ENCODES

Diuretics force a natriuresis and inflate FENa regardless of true volume status — the single most common reason FENa misleads on the wards. When diuretics are on board, FENa is uninterpretable; switch to FEUrea, which is minimally affected by loop/thiazide agents.

BOARD TRAP

A high FENa drawn after a dose of furosemide proves the patient is euvolemic/has ATN. WRONG — the diuretic forces natriuresis and inflates FENa regardless of true volume; this is the most common ward pitfall. Use FEUrea <35% to reveal prerenal physiology.

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12. Use FeUrea instead

WHAT YOU SEE

Bullets 'FeUrea <35% → prerenal; FeUrea >50% → intrinsic ATN'

WHAT IT ENCODES

FEUrea = (UUrea × PCr) / (PUrea × UCr) × 100. Because urea reabsorption is proximal and largely diuretic-independent, FEUrea stays valid on diuretics: <35% = prerenal, >50% = intrinsic/ATN. It is the go-to when FENa is contaminated by diuretic use.

BOARD TRAP

Keep trusting FENa even after starting furosemide. WRONG — once diuretics are given, FENa is falsely high; FEUrea <35% is the correct discriminator for prerenal physiology in the diuretic-exposed patient.

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13. Single most important sentence

WHAT YOU SEE

Bottom: 'FENa is not useless... just less trustworthy — use as supporting evidence, not the final answer'

WHAT IT ENCODES

Board takeaway: FENa is SUPPORTING evidence, never the sole determinant. The most informative pieces are history + exam, urine sediment (muddy-brown casts = ATN; bland/hyaline = prerenal; RBC casts = GN; WBC casts = AIN), and response to a fluid challenge. Weigh FENa within that context, especially given its many <1% intrinsic exceptions (sepsis, contrast, rhabdo, GN, HRS).

BOARD TRAP

The FENa value is the definitive test that settles prerenal vs intrinsic AKI. WRONG — it is one supportive data point with numerous exceptions; sediment, history, and fluid responsiveness override a borderline FENa.
